



ROHIT NAGAWELLY

Mumbai, India 400097
9867855572
nagawellyrohit@gmail.com

SUMMARY

Computer Science Engineer with a Bachelor’s degree from the University of Mumbai and a Master’s in Advanced Computer Science with Artificial Intelligence from the University of Strathclyde, UK. Possessing a strong foundation in Python, Machine Learning, Deep Learning, and AWS cloud technologies, along with solid expertise in Java and core computer science principles, such as data structures and algorithms. Hands-on experience in building, training, and optimising machine learning models, including Multi-Layer Perceptrons (MLPs), and developing data-driven AI solutions for real-world applications. Demonstrates strong analytical thinking, problem-solving skills, and adaptability in dynamic environments. Actively seeking GenAI developer or AI/ML Engineer, or LLM Engineer roles where advanced AI knowledge and practical engineering skills can contribute to scalable, intelligent, and impactful solutions.

SKILLS

- Flexible and Adaptable
- Critical Thinking
- Public Speaking
- Leadership
- Problem- Solving
- Time management

EXPERIENCE

TATA TECHNOLOGIES, 07/2023 - 09/2023
Intern, Mumbai, India

MACHINE LEARNING INTERN

- Assisted in developing machine learning models for real-world engineering applications.
- Performed data preprocessing, feature selection, and model training using Python.
- Evaluated model performance using accuracy metrics and validation techniques.
- Collaborated with cross-functional teams to understand practical deployment challenges.
- Gained exposure to industry-level workflows in AI and data-driven decision systems.

EDUCATION AND TRAINING

University of Strathclyde, Glasgow, UK, 2025
Master of Science in Advanced Computer Science with AI
Rizvi College of Engineering, Mumbai, India, 2024
Bachelor of Computer Engineering
Shri T.P. Bhatia College, Mumbai, India,2020 HSC
Vivek Vidyalaya, Mumbai, India, 2018 SSC

PROFESSIONAL SKILLS

HTML	MACHINE LEARNING
SQL/	DEEP LEARNING
NO SQL	DATA STRUCTURE
PYTHON	ALGORITHMS
JAVA	NLP
C++	AWS



PROJECTS

Retrieval-Augmented Generation (RAG) System – MSc Dissertation

University Of Strathclyde, Glasgow, UK

- Designed and implemented an end-to-end retrieval-augmented generation pipeline combining document retrieval and generative AI to produce accurate, context-aware answers. Built a domain-specific knowledge base by curating and preprocessing datasets from Kaggle, including text cleaning, chunking, and embedding generation. Leveraged semantic search with dense vector embeddings (using faiss) to retrieve the most relevant documents for user queries. Integrated retrieved context with a generative language model (HuggingFace Transformers) to generate precise, grounded responses, significantly reducing hallucinations common in LLM outputs. Conducted extensive experiments and evaluations to optimise retrieval quality and response accuracy using quantitative metrics and user feedback. Developed the entire system in Python, using Jupyter Notebook for prototyping and iterative testing. Documented architecture, workflow, and results in a detailed thesis, demonstrating the practical viability of RAG for real-world AI applications.
Technologies: Python, HuggingFace Transformers, Faiss, RAG, Vector Embeddings, Jupyter Notebook.

AI-Driven Autonomous Traffic Management System

- Built an intelligent traffic system to control traffic signals using real-time video feeds and reduce congestion. Used computer vision to detect vehicles, pedestrians, and traffic signals. Developed a reinforcement learning agent to optimize signal timings and reduce waiting times. Managed multiple intersections with multi-Agent RL for city-wide traffic optimization. Created a dashboard to show traffic flow and system performance. Achieved 30% faster traffic flow compared to traditional fixed-timing signals.
Technologies: Python, OpenCV, TensorFlow/PyTorch, reinforcement learning

Generative AI Document Summarizer

- Developed a system to ingest multiple document types (reports, PDFs, research papers) and generate concise, context-aware summaries using generative AI. Implemented text chunking, embedding generation, and vector-based retrieval to identify relevant content before summary generation. Achieved over 85% accuracy in summary relevance and reduced document review time by 70% in testing. Built an interactive web application for real-time summarisation and hosted the project on GitHub for demonstration.
Technologies: Python, HuggingFace Transformers, LangChain, FAISS/Chroma, OpenAI GPT, Streamlit, AWS.

IoT-Based Anti-Sleep Alarm

- Designed and developed a safety system to monitor driver alertness and prevent accidents caused by drowsiness during vehicle operation. Utilized sensors such as accelerometers and infrared cameras to track eye movements, head position, and steering patterns indicative of driver fatigue. Implemented real-time detection algorithms to analyse sensor data and identify signs of sleepiness. Integrated an alert mechanism that triggers audible alarms, vibrations, or visual warnings to prompt the driver to take necessary breaks. Developed the system using IoT microcontrollers (Arduino/Raspberry Pi) and sensor interfacing, demonstrating real-time embedded system design. Highlighted the project's importance in enhancing road safety during long-distance and night driving scenarios.
Technologies: Python/C++, Infrared Sensors, IoT Communication Protocols.

Face Mask Detector

- Developed a computer vision system to detect whether individuals are wearing face masks in images and video streams, aimed at supporting public health safety measures. Utilized deep learning techniques and convolutional neural networks (CNN) to classify images with and without masks. Reprocessed dataset with image augmentation to improve model robustness against varied lighting and angles. Achieved high accuracy in mask detection, making the system suitable for deployment in public places such as airports, hospitals, and retail stores. Demonstrated real-time detection capability using OpenCV for video stream processing.
Technologies: Python, OpenCV, TensorFlow/Keras, CNN, NumPy.